

Security+ 5.1 · Data Roles & Responsibilities

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 5.1 — Data Roles & Responsibilities (2:27)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Data roles define responsibilities: Owner (business manager — defines classification and who accesses), Controller (determines purpose and legal use — GDPR/LGPD), Processor (processes on behalf of controller — e.g., cloud provider), Custodian (IT — administers and implements owner's requirements).

High-Yield Core Concepts Tested on Exam

- Owner = business manager (defines data classification and access)
- Custodian = IT (implements and administers data protection)
- Controller (GDPR/LGPD) determines the purpose of data processing
- Processor processes data on behalf of and under instructions from the controller
- Data owner is accountable; custodian is responsible for implementation

Exam Tips & Classic Scenarios

- 'the person who determines how data is used' = data controller
- 'business manager who defines who accesses data' = data owner
- 'IT person who implements access controls' = data custodian
- 'cloud provider processing data on behalf of company' = data processor

Typical Exam Scenario: An HR manager defines who can access employee data. This is the data owner (HR). If a cloud provider processes that data on behalf of the company, it is the data processor. The exam may ask to identify the role based on the description.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 5.1 — Data Roles & Responsibilities)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. `a = 5\nb = 3\na, b = b, a\nprint(a - b)`

- A) 2
- B) -2
- C) 8
- D) 0

2. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'Owner=business manager (defines); Custodian=IT (implements)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'Owner=business manager (defines); Custodian=IT (implements)'...

3. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'Controller (LGPD) defines the WHY; Processor processes ON ORDER'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Controller (LGPD) defines the WHY; Processor processes ON ORDER...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'the person who determines how data is used' = data controller.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'the person who determines how data is used' = data controller...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'HR manager defines who accesses employee data' = data owner (HR). What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'HR manager defines who accesses employee data' = data owner (HR)
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'Owner=business manager (defines); Custodian=IT (implements)'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Owner=business manager (defines); Custodian=IT (implements)...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'Controller (LGPD) defines the WHY; Processor processes ON ORDER'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'Controller (LGPD) defines the WHY; Processor processes ON ORDER'...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'the person who determines how data is used' = data controller. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'the person who determines how data is used' = data controller.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'HR manager defines who accesses employee data'=data owner (HR)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'HR manager defines who accesses employee data'=data owner (HR)....

10. Corporate scenario (Data Roles & Responsibilities): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 5.1), an analyst evaluates the following technical requirement: 'Owner=business manager (defines); Custodian=IT (implements)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Owner=business manager (defines); Custodian=IT (implements)
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — -2.

✓ **Why it is correct:** The correct option is '-2'. The code performs a swap of values between variables 'a' and 'b' using multiple assignment in Python. Initially, a=5 and b=3. The line 'a, b = b, a' evaluates the right side first, creating a tuple (3, 5), then unpacks that tuple into the left-side variables, resulting in a=3 and b=5. Then, 'print(a - b)' computes 3 - 5, which equals -2. The option listed as correct in the answer key is index 1, which is '-2'. Therefore, the correct answer is '-2'.

✗ **Why the others are wrong:**

- 2: The option '2' represents the absolute value of the difference or the result of the inverse operation (b - a). In the context of the code, after the swap, a=3 and b=5, so a - b = -2. The value '2' would be obtained if the subtraction were b - a (5 - 3 = 2) or if the modulus of the difference were calculated. Since the expression is 'a - b', the result is negative. In information security, confusing the order of operands in verification calculations (such as checksums or hashes) can lead to false positives or negatives, but here it is a simple arithmetic error. The option is wrong because it ignores the negative sign resulting from the order of subtraction after the swap.
- 8: The option '8' would correspond to the sum of the original values (5 + 3 = 8) or string concatenation, but it is not the result of the subtraction after the swap. In programming, addition is a different operation from subtraction. In security, confusing arithmetic operations can lead to vulnerabilities such as buffer overflow or logic errors in cryptographic algorithms, but here it is just a calculation error. The option is wrong because it does not perform the subtraction operation as per the code.
- 0: The option '0' would be the result if the subtraction were between equal values, which does not occur after the swap (a=3, b=5). In security, a zero result can indicate success in comparisons (such as in memcmp), but here it does not apply. The option is wrong because the difference between 3 and 5 is not zero.

2) D — Implement the corresponding solution: 'Owner=business manager (defines); Custodian=IT (implements)'....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): Owner=business manager (defines); Custodian=IT (implements).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) A — Implement the corresponding solution: Controller (LGPD) defines the WHY; Processor processes ON ORDER....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): Controller (LGPD) defines the WHY; Processor processes ON ORDER. Under data protection regulations like LGPD and GDPR, the controller determines the

purposes and means of processing personal data, while the processor acts on behalf of the controller and processes data only upon the controller's instructions.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

4) B — Implement the corresponding solution: 'the person who determines how data is used' = data controller....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): 'the person who determines how data is used' = data controller. The data controller is responsible for determining the purposes and means of processing personal data, as defined in regulations like GDPR.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: 'HR manager defines who accesses employee data' = data owner (HR).

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): 'HR manager defines who accesses employee data' = data owner (HR). The data owner is a senior executive responsible for the data's classification, protection, and access decisions. In this case, the HR manager acts as the data owner for employee data, determining who can access it.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and it breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) A — Implement the corresponding solution: Owner=business manager (defines); Custodian=IT (implements)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): Owner=business manager (defines); Custodian=IT (implements).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

7) B — Implement the corresponding solution: 'Controller (LGPD) defines the WHY; Processor processes ON ORDER'....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): Controller (LGPD) defines the WHY; Processor processes ON ORDER.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) B — Implement the corresponding solution: 'the person who determines how data is used' = data controller..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): 'the person who determines how data is used' = data controller. The data controller is the entity that determines the purposes and means of processing personal data, bearing primary responsibility for compliance with privacy regulations such as GDPR.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'HR manager defines who accesses employee data'=data owner (HR).....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): 'HR manager defines who accesses employee data'=data owner (HR).

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) B — Implement the corresponding solution: Owner=business manager (defines); Custodian=IT (implements).

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 5.1 (Data Roles & Responsibilities): Owner=business manager (defines); Custodian=IT (implements). The data owner is accountable for classification and policy, while the custodian handles day-to-day protection.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.